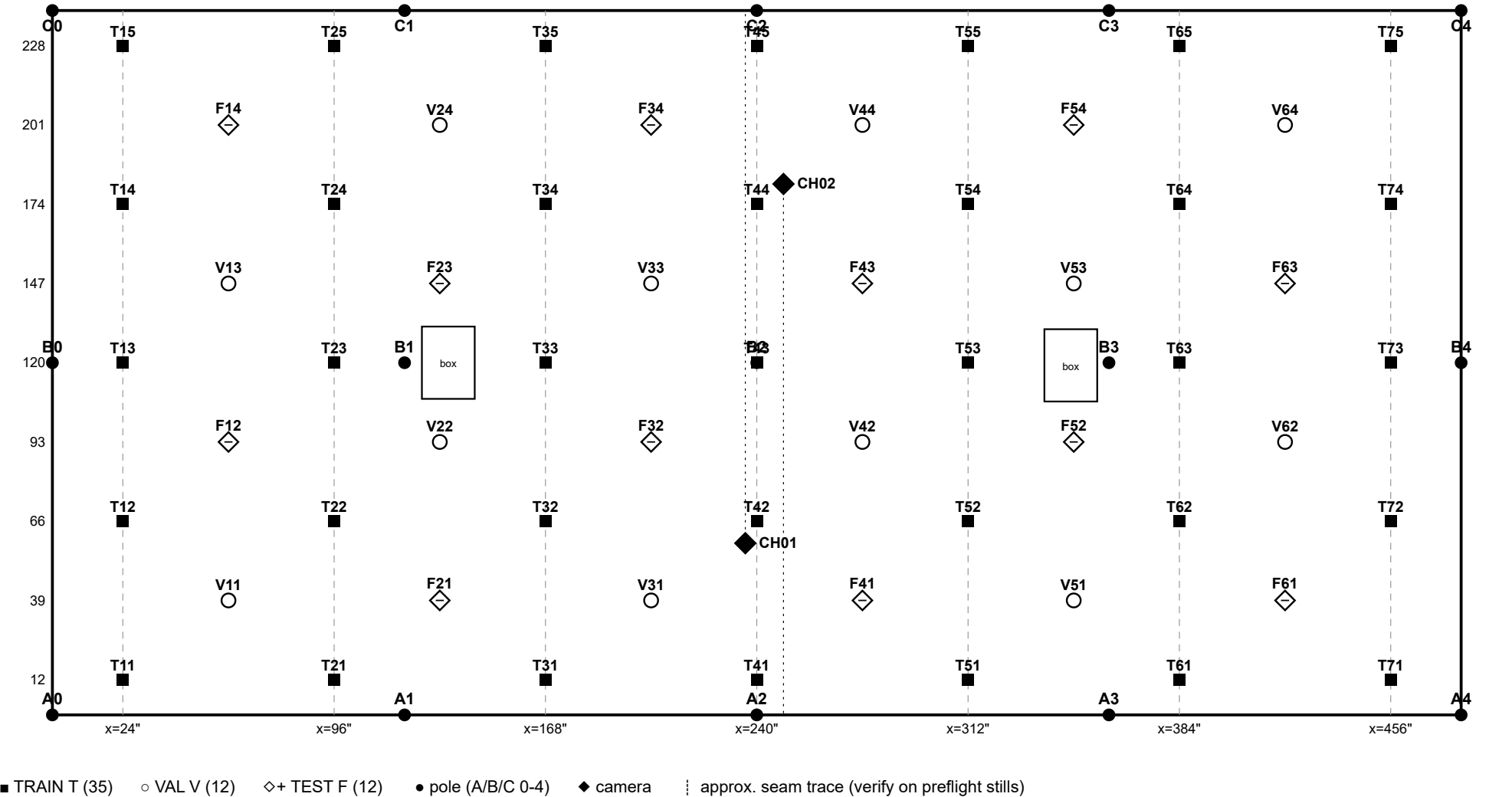


CH01/CH02 Calibration - Stakeout Map (PLAN_CH01_CH02_pano_ground_map.md v2 · board: ChArUco 12×9 @60 mm, DICT_5X5_100, 72 cm long edge along +x

· ALL FIELD COORDINATES IN INCHES; pipeline converts ×2.54 to cm)

Date:	Operator:	Weather / light:	Phone-PC clock offset: ____ s
5-square span, x-dir: ____ mm (expect 300)	y-dir: ____ mm	Board thickness: ____ mm	Typical grass height: ____ in
Field long edge 1: ____ in (expect 480)	Long edge 2: ____ in	Diagonal 1: ____ in (expect ~536.7)	Diagonal 2: ____ in
Cord height above ground (typ.): ____ in	Train cords: flags @ 12/66/120/174/228 in pre-taped √: ____	VAL/TEST cords: flags @ 39/93/147/201 in pre-taped √: ____	



Stakeout flow (checkbox by default - no per-station taping): ① indoors: tape 5 flags on each of the 7 train cords @ 12/66/120/174/228 in, measured from the cord's wall-A zero mark under deployment tension (measure once - all cords identical) ① on site: tape each cord's x from the corners along BOTH long walls, mark the wall ends (table below) ② stretch the cord taut at soil level,

wall-A zero mark at the wall face, **cones weighting both ends** ③ place the board: **54 cm short edge hugging the cord, origin corner at the flag**, +x arrow toward the far end (A4) → position + orientation aligned in one motion → checkbox ④ only a board that cannot land at its flag (obstruction) gets moved and hand-measured (the * column) ⑤ do not block any camera's line of sight to the board (step 2-3 m aside; being in frame is harmless), hold 8-10 s. **Only cross-width cords are used** - flags carry y, so length-wise cords add nothing. **IDs are paper bookkeeping only** (T34 = cord 3, flag 4); no ID marks on the field, cameras never need to see them - identification = time + order + position self-consistency. **Cones:** weight cord ends + mark upcoming stations for wayfinding; never a precision datum (a cone base is ±several cm ambiguous). **Iron rules:** exactly ONE board on the field at any time · everything matte, NO retroreflective tape (glare / IR blooming kills nearby corner detection) · SEAM rows are placed after checking preflight stills · call remote QC after each batch before continuing · before leaving: wait for _to_ closure, verify + back up · ground stations serve CH03/04 too (all cameras record together).

Cord wall-end measurements (the construction chain's foundation - two numbers per cord)

Cord	Target x (in)	Wall-A end: measured x	Wall-C end: measured x	Taut + flags OK	Notes
L1	24				
L2	96				
L3	168				
L4	240				
L5	312				
L6	384				
L7	456				

VAL/TEST cords (x = 60, 132, 204, 276, 348, 420 in) same method, flags @ 39/93/147/201 in. Cords stay down all session (straightness validation); cones weight the ends. NO length-wise cords - the flags on the cross cords already carry y.

CH03/04 intrinsics sweep (during waits between placement batches)

Camera	Start	End	Valid poses (target 20-30)	Frame edges covered?	QC ok	Notes
CH03						
CH04						

Sweep: hand-hold the board in front of the camera varying distance / position / tilt (±30-45°), holding each pose still 1-2 s; CH03 first → remote QC confirms detection quality → then CH04. Avoid sun-glare angles.

Log 1/2 - TRAIN T11-T75 (35) + start of VAL/TEST

ID	Set	Target x (in)	Target y (in)	On cord @ flag ✓	Held 8-10 s	QC ok	Measured x,y (moved only)*	Notes
T11	TRAIN	24	12					
T12	TRAIN	24	66					
T13	TRAIN	24	120					
T14	TRAIN	24	174					
T15	TRAIN	24	228					
T21	TRAIN	96	12					
T22	TRAIN	96	66					
T23	TRAIN	96	120					
T24	TRAIN	96	174					
T25	TRAIN	96	228					
T31	TRAIN	168	12					
T32	TRAIN	168	66					
T33	TRAIN	168	120					
T34	TRAIN	168	174					
T35	TRAIN	168	228					
T41	TRAIN	240	12					
T42	TRAIN	240	66					
T43	TRAIN	240	120					
T44	TRAIN	240	174					
T45	TRAIN	240	228					
T51	TRAIN	312	12					
T52	TRAIN	312	66					
T53	TRAIN	312	120					
T54	TRAIN	312	174					
T55	TRAIN	312	228					
T61	TRAIN	384	12					
T62	TRAIN	384	66					
T63	TRAIN	384	120					
T64	TRAIN	384	174					
T65	TRAIN	384	228					
T71	TRAIN	456	12					
T72	TRAIN	456	66					
T73	TRAIN	456	120					
T74	TRAIN	456	174					
T75	TRAIN	456	228					
V11	VAL	60	39					

Set rule: V/F alternate in a checkerboard pattern, pre-assigned - never reassign on site; a moved station keeps its set, only its coordinates change.

Log 2/2 - VAL/TEST (cont.) + SEAM S1-S6 + re-shoot/moved X1-X6

ID	Set	Target x (in)	Target y (in)	On cord @ flag ✓	Held 8-10 s	QC ok	Measured x,y (moved only)*	Notes
F12	TEST	60	93					
V13	VAL	60	147					
F14	TEST	60	201					
F21	TEST	132	39					
V22	VAL	132	93					
F23	TEST	132	147					
V24	VAL	132	201					
V31	VAL	204	39					
F32	TEST	204	93					
V33	VAL	204	147					
F34	TEST	204	201					
F41	TEST	276	39					
V42	VAL	276	93					
F43	TEST	276	147					
V44	VAL	276	201					
V51	VAL	348	39					
F52	TEST	348	93					
V53	VAL	348	147					
F54	TEST	348	201					
F61	TEST	420	39					
V62	VAL	420	93					
F63	TEST	420	147					
V64	VAL	420	201					
S1	SEAM*							
S2	SEAM*							
S3	SEAM*							
S4	SEAM*							
S5	SEAM*							
S6	SEAM*							
X1	EXTRA							
X2	EXTRA							
X3	EXTRA							
X4	EXTRA							
X5	EXTRA							
X6	EXTRA							

* Measured columns only when the board could not land at its flag; "On cord @ flag" checked = coordinates are the target values. SEAM rows: locate the actual seam on preflight stills first; one group each near / mid / far, sample BOTH sides, coordinates hand-measured. QC ok = remote frame-pull confirmed usable corners.

Survey (must measure before teardown) - poles & cameras CH01-CH06

Pole	Design x (in)	Design y (in)	Measured x	Measured y	Notes
A0	0	0			
A1	120	0			
A2	240	0			
A3	360	0			
A4	480	0			
B0	0	120			
B1	120	120			
B2	240	120			
B3	360	120			
B4	480	120			
C0	0	240			
C1	120	240			
C2	240	240			
C3	360	240			
C4	480	240			

Channel	Mount description (pole/wall)	Measured x	Measured y	Lens height (in)	Aim / notes
CH01					
CH02					
CH03					
CH04					
CH05					
CH06					

CH07/08 (in-box) deferred by operator decision; box positions are recoverable from video. **But interior clear dims + wall thickness are NOT recoverable from video** - tape-measure the boxes before disposal (keep them after teardown, or measure-then-discard; 5 minutes).